# Quality Assessment of Multispectral Remote Sensing Images: A Subjective Study and No Reference Algorithms

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***Abstract*—Multispectral remote sensing images suffer from a wide range of degradations due to noise, blur, haze, compression, striping, and other distortions. While there is rich literature in the restoration of such multispectral images, the quality assessment (QA) of multispectral images has received much less attention. In addition, often a reference image is not available, which motivates the study of no-reference (NR) image QA (IQA). While several algorithms for NR-IQA have been designed for natural images, their relevance for multispectral images has not been validated. The major impediment to achieving this is the lack of a subjective QA database for validation. We make three contributions in our work. We first design a multispectral IQA database, namely, the multispectral remote sensing image database (MRSID), with images suffering from various distortions and conduct a subjective study to obtain pairwise image quality rankings. We then benchmark several NR-IQA methods developed for natural images on MRSID. We show that recent vision-language-based models can yield excellent performance when fine-tuned on our dataset, especially when multispectral images are trained with images pooled from all bands, achieving an improvement of 15.8% in pairwise preference accuracy over single-band training. Finally, we investigate the reasons for such superior performance and show that improving the alignment of features across different spectral bands can further improve performance, yielding an additional 3.1% gain in pairwise preference accuracy on the cross-scene evaluation split, indicating improved generalization.**

***Index Terms*—Benchmark, deep learning, distortion, multispectral imaging image quality, quality assessment (QA), remote sensing, subjective study.**

## I. Introduction

MULTISPECTRAL images captured using satellites suffer from a variety of quality degradations such as noise, blur, compression, striping, and haze. In particular, stripes and banding in satellite images occur due to nonuniform detectors in the charge-coupled device (CCD) array. Images are blurred due to lens optics, detector sampling, sensor electronics, atmospheric scattering, and turbulence of the satellites. Images are corrupted by noise due to errors in the measurement of photons collected. In addition, the image quality is also degraded due to the presence of haze and clouds in the scene. The automatic assessment of the quality of such images is extremely important in providing a framework for automatically analyzing and correcting for these distortions. Moreover, we seek to assess quality bandwise to provide tools to monitor and control quality in individual bands.

Although there are several algorithms for the restoration of multispectral images, the quality assessment (QA) of the corrupted and restored images has not been studied extensively. Furthermore, in several of these applications, a reference image of high quality is not available for comparison to evaluate the quality of the distorted image. This motivates the study of no-reference (NR) image QA (IQA) methods that operate without the availability of a reference image. Traditional metrics such as SNR [1] and MTF [2] are computed over carefully selected homogeneous patches or sharp edges, often ignoring degraded or artifact-rich regions. In contrast, models trained on large-scale datasets can yield automated, scene-level perceptual quality scores that reflect overall image usability, enabling evaluation in heterogeneous scenes as well. While both classical and deep-learning-based models have become quite popular for natural images, their relevance for multispectral remote sensing images is still not clear for multiple reasons.

Benchmarking existing NR-IQA methods and evaluating newer algorithms for distorted multispectral and panchromatic remote sensing imagery requires a dataset that includes subjective quality annotations across diverse remote sensing images, particularly in the absence of reference images. Quality models trained on such datasets can benefit downstream applications by weeding out degraded images. For example, in urban planning and infrastructure mapping, degraded images may lead to inaccurate structural details. In surveillance tasks that include object detection and vehicle tracking, localized blur or noise can severely hinder performance. Recent advances leverage hyperspectral and heterogeneous remote sensing data for improved land cover change detection [3], [4] though distortions in multispectral images can degrade detection accuracy. In solar farm health monitoring and inspection,

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haze or striping can obscure panel edges or mask thermal anomalies. In addition, such quality models enable more reliable benchmarking of restoration algorithms designed to improve image quality.

Several IQA databases for remote sensing focus on pansharpened IQA. The pansharpened remote sensing image quality database (PRSIQD) [5] is a large-scale repository of pansharpened images from diverse satellite sensors, annotated with human subjective scores. Perceptual studies on pansharpened images have also explored the link between fusion methods and perceived visual quality [6]. Other datasets in remote sensing IQA focus on SAR images [7], synthetic distortions for panchromatic images [8], and UAV hyperspectral images [9]. Existing studies do not contain rich real distortions in multispectral bands with large-scale subjective annotations. Some studies on the visual quality of native remote sensing imagery rely primarily on limited expert visual inspection and standard objective metrics [10], [11], [12], [13]. Existing remote sensing IQA datasets either focus on pansharpening, SAR evaluation, or UAV hyperspectral imaging. They typically operate on small-scale or expert-only subjective data or only synthetic distortions for multispectral images. Our multispectral remote sensing image database (MRSID) helps fill a gap in publicly available datasets that provide large-scale, systematically collected human quality judgments for multispectral satellite images spanning a diverse set of real-world distortions.

Popular classical supervised NR-IQA methods [14], [15] are trained on natural images. However, their effectiveness for multispectral images remains unclear. Second, unsupervised methods such as NIQE [16] rely on the use of a corpus of pristine natural images, which may not be appropriate for multispectral remote sensing images. Deep learning-based IQA methods [17], [18], [19], [20], [21], [22], [23], [24] trained on a large number of natural images may also suffer from a significant domain shift when these models are applied on multispectral images.

One of the major limitations that prevent us from validating the relevance of existing IQA methods for multispectral images is the lack of a subjective IQA database. Second, it is not clear whether improvements in NR-IQA designs for natural images translate to corresponding improvements in multispectral images. Third, the question of how to use multiple spectral bands while training or fine-tuning NR-IQA methods has been unexplored. Specifically, it is not known whether there are benefits in training a unified model across different spectral bands in comparison to training separate models for individual bands. Moreover, the role of effective use of multiple spectral bands has also not been studied much in the context of designing NR-IQA methods. Our work focuses on addressing the above challenges in NR-IQA of multispectral remote sensing images.

Our first contribution involves creating a large database of multispectral images suffering from a wide variety of distortions. In particular, we create a database consisting of 1101 multispectral and panchromatic images, namely, the MRSID, suffering from both real-world and synthetic distortions. The images are drawn from WorldView-2, WorldView-3 [25], and Cartosat-3 [26] satellites. We then conduct a subjective study to obtain human opinions for ranking 3000 image pairs. This leads to a total of 16 518 subjective rankings obtained through our subjective study.

We then benchmark several existing NR-IQA methods on our database in various settings by fine-tuning these models with the pairwise rankings available in our database. We observe that deep learning-based methods offer significantly higher performance than the classical NR-IQA methods on multispectral images. Second, we find that fine-tuning NR-IQA methods jointly on multiple spectral bands to obtain a unified model significantly improves performance compared to fine-tuning different models on individual bands. Furthermore, we find that recent vision-language models based on visual question answering (Q-Former [22]) yield state-of-the-art performance compared to other methods in the literature.

Upon investigating the reasons for such superior performance, we find that these vision-language models tend to align the features better across different spectral bands compared to other models. This also allows one to leverage more data across bands and improve performance. Thus, we impose a centered kernel alignment (CKA)-based loss to enable the features learned across spectral bands to align better. We show that imposing this leads to better alignment of the learned features, leading to further improvement of their performance in ranking the quality of multispectral image pairs.

In summary, our main contributions in this work are given as follows.

1) We create a database of 1101 multispectral and panchromatic images suffering from both realistic and synthetic distortions.
2) We conduct a pairwise subjective study and collect 16 518 human rankings for 3000 image pairs.
3) We benchmark existing classical and deep learning-based NR-IQA algorithms on our database in diverse settings and show that a unified model trained jointly on multispectral bands can yield superior performance compared to individual models trained on specific spectral bands.
4) We investigate the reasons for the superior performance of vision-language models and further improve their performance by better aligning the features from different spectral bands.

Our database and code will be made publicly available at *github.com/Aakankshabharti/mrsid_qa.git*. The rest of this article is organized as follows. We survey related work in Section II and present our database and subjective study in Section III. A detailed account of benchmarking of existing NR-IQA measures is provided in Section IV. In Section V, we analyze why the Q-Former model performs better while training jointly with multiple spectral bands and improve its performance further. We finally conclude this article in Section VI.

## II. Related Work

We first survey related works on subjectively annotated datasets for IQA of remote sensing images. We then survey

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related work in NR-IQA of natural images using both classical and deep learning-based methods. We also survey related work in the QA of pansharpening of multispectral images.

### A. Datasets for QA of Remote Sensing Images

Existing subjective IQA datasets for remote sensing imagery have largely targeted specific modalities or processing tasks. There are numerous works focused on pansharpened image evaluation. The PRSIQD dataset [5], for instance, comprises roughly 7000 pansharpened WorldView-2 images annotated with mean opinion scores (MOSs) from 15 human observers, enabling evaluation of spatial–spectral tradeoffs in fusion algorithms. Similarly, Agudelo-Medina et al. [6] conducted a subjective quality study involving 33 subjects who rated 420 pan-sharpened images with typical distortions such as blur and noise, providing valuable perceptual ground truth for pansharpened IQA. Vivone et al. [27] proposed a benchmarking protocol and dataset (PAirMax) for pansharpening consisting of 14 PAN and multispectral image pairs; however, it lacks subjective ground truth. In the SAR domain, the SSDD ship detection dataset is annotated with expert-labeled three usability categories from usable to unusable, covering over 1100 images from sensors such as RadarSat-2, TerraSAR-X, and Sentinel-1, with labels reflecting operational detectability [7]. Usability-based IQA datasets [28] annotate remote sensing images with binary or ordinal labels reflecting interpretability in practical workflows. However, existing subjective datasets remain largely confined to pansharpened or nonsatellite scenarios.

Expert visual examinations have identified key distortions affecting remote sensing imagery. For example, Agudelo-Medina et al. [6] visually analyzed blur and additive noise in pansharpened outputs, assessing how these degrade perceptual quality and developing corresponding QA models. Visual inspection has also been applied to quality inspection of land-use survey images, where distortions such as haze and striping were reported as significant challenges [10]. However, most studies remain limited to a small set of images and distortions, highlighting the need for a broader analysis across sensors and imaging conditions.

### B. Classical NR-IQA Methods

Most successful NR-IQA methods are based on the use of features derived from natural scene statistics (NSSs). BRISQUE [14] is based on NSS extracted in the spatial domain and has been widely used due to its simplicity and effectiveness. Other methods such as DIIVINE [15] and BLIINDS [29] employ NSS features extracted from steerable pyramids and discrete cosine transform, respectively. CORNIA [30] constructs quality-aware visual codebooks through unsupervised learning to design effective features for NR-IQA. HOSA [31] improvises on CORNIA and computes higher-order statistics of local image descriptors using codebooks to quantify perceptual quality. On the other hand, NIQE [16] is an opinion-unaware model that estimates quality using a multivariate Gaussian model fit on the NSS features used in BRISQUE. The features of a test image are compared with those from a pristine corpus, eliminating the need for subjective score training. IL-NIQE [32] builds upon NIQE by aggregating local patch-level statistics, enhancing sensitivity to localized image degradation.

### C. Deep Learning-Based NR-IQA Methods

Recent deep learning approaches achieve significantly superior NR-IQA performance for natural images compared to handcrafted features. Initially, convolutional neural network (CNN)-based approaches were successfully exploited to design several QA methods. DBCNN [17] employs a dual-stream architecture that combines pathways designed for capturing synthetic and authentic distortions via bilinear pooling. HyperIQA [18] learns hierarchical representations and adaptively weighs multilevel features to better model perceptual quality. MetaIQA [24] utilizes metalearning strategies for fast adaptation to unseen distortion types or new domains, increasing its generalization ability. On the other hand, transformer-based architectures have also been successfully applied in NR-IQA. MUSIQ [23] processes images at multiple scales to capture both fine-grained and global distortion characteristics. TReS [20] leverages transformer architectures to learn nonlocal representations in conjunction with CNN architectures to extract local features. MANIQA [19] incorporates multidimensional attention mechanisms that focus on distortion-relevant regions while preserving semantic content.

While the above models focus largely on supervised training, CONTRIQUE [33] exploits self-supervised learning of image features to learn rich quality representations. ReIQA [21] improvises on CONTRIQUE to separately capture low-level (quality-sensitive) and high-level (content-aware) representations, achieving perceptually aligned predictions. More recently, vision-language models have also been successfully used for NR-IQA. CLIP-IQA [34] shows promising zero-shot performance, while LIQE [35] trains the text encoder to show robust performance on diverse distortions. Q-Former [22] exploits the rich pretraining of visual question answering models and fine-tunes them for NR-IQA to show excellent generalization performance across diverse distortions.

### D. IQA on Remote Sensing Images

While the problem of generic multispectral IQA has not been studied much in the literature, the particular case of pansharpened images has been studied quite extensively. Pansharpened IQA can be broadly classified into full-reference (FR) and NR approaches, similar to natural IQA literature. FR methods evaluate the quality of a pansharpened image by comparing it with a reference high-resolution multispectral (HRMS) image. Common FR metrics include ERGAS [36], SAM [37], UIQI [38], and $Q2^n$ [39]. In contrast, NR approaches are designed for scenarios where a reference HRMS image is unavailable. Prominent NR methods include QNR [40], Hybrid QNR [41], and the method proposed by Khan et al. [42]. These algorithms estimate quality by separately measuring spectral and spatial distortions. For instance, QNR uses bandwise similarity with the PAN image

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before and after fusion to estimate spatial distortion and interband similarity changes to estimate spectral distortion. More recently, learning-based no-reference methods [43] leverage deep feature extraction from PAN, LRMS, and pansharpened images to approximate reference features and predict quality scores without requiring ground-truth HRMS images. A recent work, DFSM-Net [44], uses Siamese deep feature similarity to measure spatial and spectral distortion in pansharpened images.

On panchromatic sensor remote sensing images, Badal et al. [8] showcase a deep feature map-driven no-reference QA method for satellite images, where FR measures were used as pseudo-labels for quality; however, the method is not evaluated on real distortions. In the SAR domain, Huang et al. [7] propose an IHHO-XGBoost hybrid model optimized via the Harris Hawk algorithm, trained on the dataset mentioned earlier. Notably, most of these methods are evaluated on pansharpened or SAR-specific datasets, with limited work on multispectral distortions.

## III. Dataset and Subjective Study

### A. Dataset Construction

Our MRSID consists of multispectral and panchromatic data from WorldView-2, WorldView-3, and Cartosat-3 sensors. We gather the WorldView satellite images from the SpaceNet Challenge datasets [25] and procure Cartosat-3 satellite images from ISRO's Bhoonidhi portal [26]. The WorldView images include scenes from Atlanta, Khartoum, Shanghai, and Vegas, whereas the Cartosat images include scenes from Rome, Thane, and Varanasi.

The multispectral scenes from the WorldView sensors contain eight bands, whereas those from the Cartosat sensors contain only four bands. To remain consistent with both sensors, we select only those bands from WorldView sensors that are present in Cartosat as well. Our dataset includes blue (B), green (G), red (R), and near-infrared (NIR) bands in multispectral scenes from both satellite sensors. The scenes cover water, desert, buildings, urban areas, green vegetation, water bodies, and roads.

The Cartosat-3 multispectral sensor provides multispectral imagery at an approximate spatial resolution of 1.12 m, while its panchromatic imagery achieves a finer resolution of 0.28 m. Similarly, WorldView-3 offers multispectral data at 1.2-m resolution and panchromatic data at 0.3 m. Corresponding to these resolutions, the Cartosat-3 multispectral scenes typically comprise around 20 000 × 20 000 pixels. In comparison, WorldView-3 scenes vary in size: those covering Khartoum, Shanghai, and Las Vegas are 4096 × 4096 pixels, while the Atlanta scene is considerably larger at approximately 14 000 × 9000 pixels. We used three multispectral scenes and one panchromatic scene from Cartosat captured regions, and we used six scenes from Atlanta, 40 scenes from Khartoum, 55 scenes from Shanghai, and 15 scenes from Vegas to extract the patches corresponding to the WorldView sensors.

From the multispectral scenes of Cartosat, we extract 500 × 500 nonoverlapping patches to get approximately 1084 patches from each Cartosat multispectral scene for each band. Similar to the multispectral scenes, we extract nonoverlapping patches of size 500 × 500 from the Cartosat panchromatic scenes to obtain 4528 patches. In the WorldView scenes, we initially extract about 16 000 patches of size 500 × 500 from the multispectral data for each band and around 16 000 patches of the same size from the panchromatic data.

We provide color composite visualizations of multispectral images in Fig. 1. The figure illustrates the standard three-channel color image (RGB) and two false-color composites following the (NIR, R, G) → (R, G, B) and (NIR, G, B) → (R, G, B) schemes, which are commonly used in the literature to highlight vegetation and water regions [45], [46], [47]. We show images that exhibit distortions in each row. Row (a) exhibits localized noise in multispectral images, which becomes more pronounced in composites where both green and blue channels are present. Row (b) shows blurring in the RGB composite, while introducing the NIR band provides higher contrast in vegetation regions, thereby reducing the perceived blur. Finally, in Row (c), the RGB composite shows slight blue hazing that reduces visibility since short-wavelength blue light is strongly scattered by the atmosphere, whereas the NIR band enhances scene details as it experiences minimal scattering and penetrates haze more effectively.

We seek to select a subset of patches that span different types of distortions and different levels of distortion within each type. Toward this, we scan through the large corpus of patches and first identify the type of distortions. The distortion types include blur, noise, haze, stripes, banding, and saturation. Since scanning through 16 000 extracted patches for each band from the WorldView sensors is not feasible, we uniformly sample 10% of the patches from each region of the WorldView scenes. The Cartosat-3 scenes were generated by applying only the standard correction levels used during the in-orbit calibration phase. This provided us with an opportunity to capture inherent real-world distortions, often resulting in patches that exhibit not only individual distortion types but also simultaneous occurrences of multiple distortions. Consequently, the patches were categorized on the basis of the perceived dominant distortion. Furthermore, based on the perceived level of distortion, we subclassify the patches within a distortion type into three coarse levels: *low, medium*, and *high*. Finally, we manually select 1101 patches from these classified patches such that each band contains a good spread of distortion types and levels.

We refer to the images described above, which were affected by distortions during scene capture, as "real distortions." We show a sample set of images affected by real distortions in each band in Fig. 2. To have more control over the type and range of distortions in multispectral images, we construct a "synthetic/simulated distortions" set by applying synthetic distortions on the clean satellite images. In Section III-B, we describe how we construct the synthetic dataset in detail.

### B. Simulated Distortions

We generate images with simulated distortions by identifying clean images from the cropped patches. We simulate diverse distortions such as Poisson–Gaussian (PoGa) noise,

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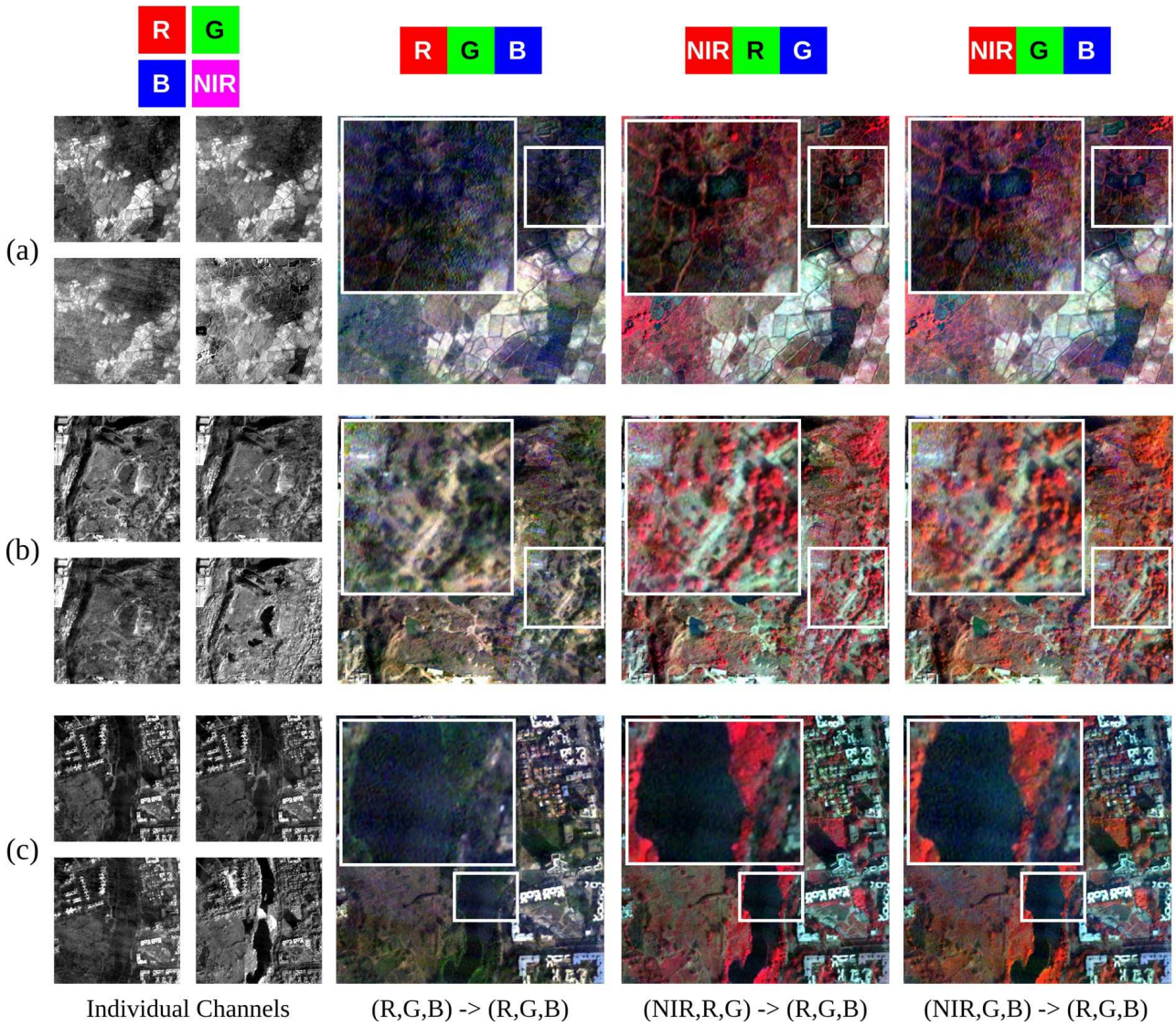



Fig. 1. Color composites derived from the multispectral images. (a) Four individual spectral bands (R, G, B, and NIR) arranged in a grid. (b) Standard three-channel color image. (c) False-color composites generated by mapping (NIR, R, and G) and (NIR, G, and B) into the three channels. Zoomed-in patches are provided at the top-left corner of each image to visualize multispectral distortions.

TABLE I

VALUES OF $a$ AND $b$ USED TO SIMULATE POGA NOISE, WHERE $y(t)$ IS POGA NOISE DEGRADED IMAGE AS $y(t) \sim \mathcal{N}(x(t), ax(t) + b)$

| Parameter | Level-1 (Low) | Level-2 (Moderate) | Level-3 (High) |
|---|---|---|---|
| $a$ | $5 \times 10^{-3}$ - $7.5 \times 10^{-3}$ | $1 \times 10^{-2}$ - $3 \times 10^{-2}$ | $3.5 \times 10^{-2}$ - $4.9 \times 10^{-2}$ |
| $b$ | $4 \times 10^{-5}$ - $7.5 \times 10^{-5}$ | $1 \times 10^{-4}$ - $2 \times 10^{-4}$ | $3 \times 10^{-4}$ - $4.5 \times 10^{-4}$ |

Gaussian-impulse noise, blur, stripes, and haze. These distortions were chosen based on empirical observations of real degradation patterns in high-resolution satellite imagery, as well as recent studies on denoising, artifact analysis, and haze removal [48], [49], [50], [51]. Together, they span both sensor-induced degradations (e.g., noise, stripes, and electronics faults) and scene-dependent effects (e.g., blur and haze). The simulated distortions are generated as described in the following.

*1) PoGa Noise:* We follow [52] to model PoGa noise as a signal-dependent noise, which has the form $y(t) = x(t) + \sigma(x(t))\zeta(t)$, where $t$ denotes the pixel coordinates, $y(t)$ is the observed image, $x(t)$ is the pristine image, $\zeta(t)$ is the zero-mean Gaussian noise with unit variance, and $\sigma(\cdot)$ is a function of $x(t)$ that gives the standard deviation of the overall-noise component. We simulate this noise independently across bands in multispectral images, as proposed in [53] and [54]. We sample the PoGa noise degraded image from $y(t) \sim \mathcal{N}(x(t), ax(t) + b)$, where $a > 0, b \geq 0$ are parameters to simulate the desired level of noise as described in [52]. We use the values presented in Table I for the parameters $a$ and $b$ to generate PoGa noise at three different scales.

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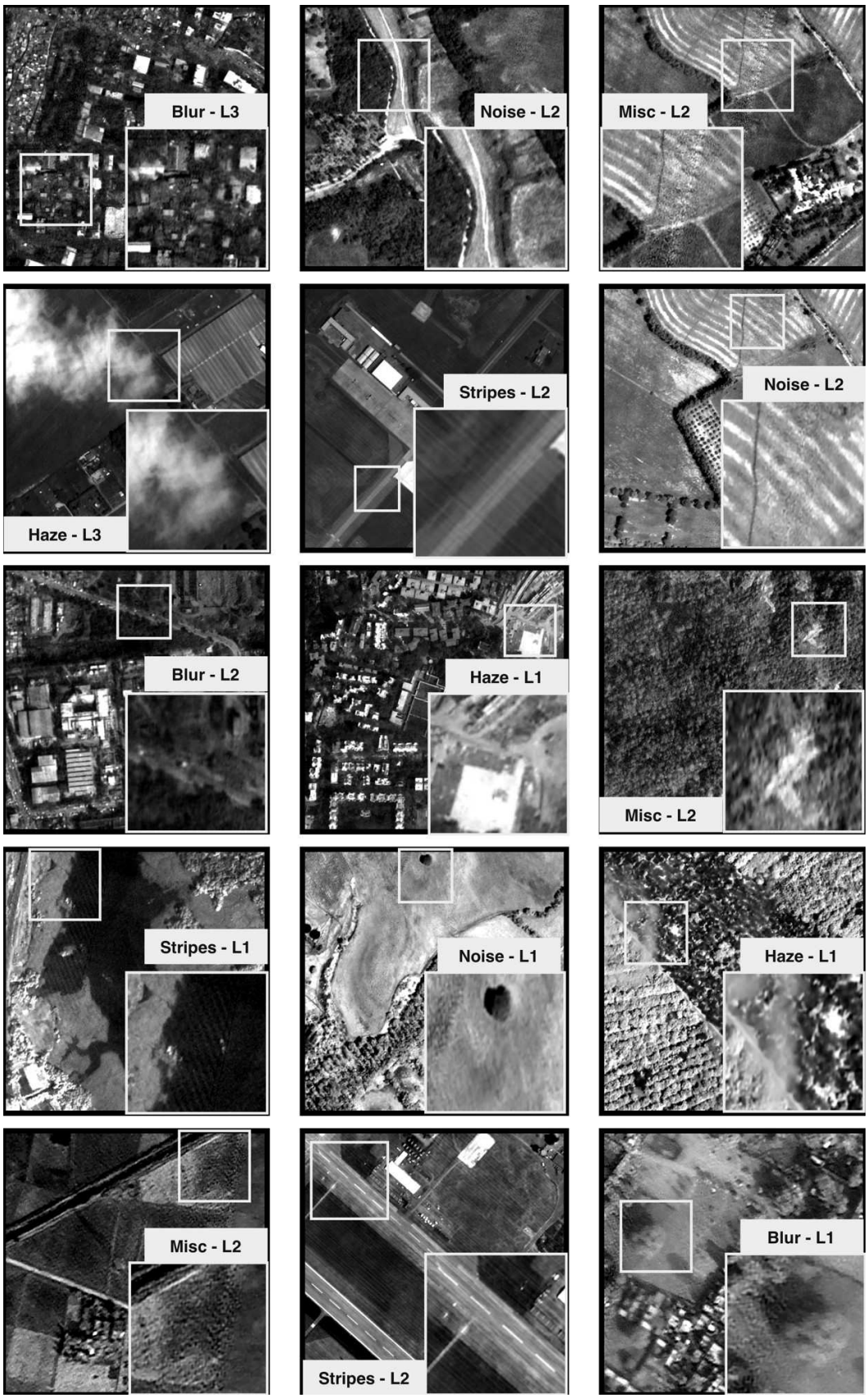



Fig. 2. Examples of real distortions from the MRSID dataset. The first four rows correspond to the multispectral bands (blue, green, red, and NIR) and the final row to panchromatic images. Each grid cell shows a distorted image with a zoomed-in region (red box) highlighting artifacts and a label specifying distortion type and level. L1, L2, and L3 denote low, medium, and high distortion levels, respectively.

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TABLE II
VALUES OF $\sigma$ AND $p$ USED TO SIMULATE GAUSSIAN-IMPULSE NOISE, WHERE $\sigma$ CORRESPONDS TO THE STANDARD DEVIATION OF THE GAUSSIAN NOISE AND $p$ CORRESPONDS TO THE PROBABILITY OF IMPULSE NOISE

| Parameter | Level-1 (Low) | Level-2 (Moderate) | Level-3 (High) |
|---|---|---|---|
| $\sigma$ | $0.03 - 0.04$ | $0.05 - 0.08$ | $0.095 - 0.12$ |
| $p$ | $0.03 - 0.04$ | $0.05 - 0.08$ | $0.095 - 0.12$ |

*2) Gaussian-Impulse Noise:* We follow the approach described in [55] and [56] to simulate additive Gaussian-impulse noise. Initially, the image is distorted by Gaussian noise, and impulse noise is added to this distorted image. Let $x_n(t)$ be the Gaussian noise distorted signal obtained as $x_n(t) = x(t) + \epsilon(t)$ where $x(t)$ is the clean image and $\epsilon(t) \sim \mathcal{N}(0, \sigma)$. The image simulated by Gaussian-impulse noise is given by $y(t) = x_n(t)(1 - b(t)) + n_{\text{imp}}(t)b(t)$, where $b(t)$ takes either 0 or 1 with probabilities $1 - p$ and $p$, respectively, and $n_{\text{imp}}(t)$ denotes the impulse noise component characterized by the uniform distribution in the range $[0.025, 0.45]$ for multispectral images and $[0, 0.5]$ for panchromatic images taking intensity values in $[0, 1]$. The values of the parameters $\sigma$ and $p$ are shown in Table II.

*3) Localized Blur:* The blur degradation process is modeled as $y(t) = b(t) * x(t) + n(t)$, where $y(t)$ is the observed image, $x(t)$ is the clean image, $b(t)$ corresponds to the blur kernel, $n(t)$ is Gaussian noise, and * represents the convolution operator [57]. To achieve realistic blur simulation, we combine three types of blur: Gaussian, defocus, and motion blur. We use the Gaussian, defocus, and motion blur kernels to generate a combined blur kernel, capturing mixed characteristics from all three types. A globally blurred image is then produced by convolving a clean image with this combined blur kernel. The kernels used are provided in the following.

1) *Gaussian Blur:* It models atmospheric turbulence, and the kernel is given as

$$h(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \tag{1}$$

2) *Defocus Blur:* It replicates the blurring effect seen in out-of-focus images and modeled as

$$h(x, y) = \begin{cases} 0 & \text{if } \sqrt{x^2 + y^2} > r \\ \dfrac{1}{\pi r^2} & \text{if } \sqrt{x^2 + y^2} \leq r. \end{cases} \tag{2}$$

3) *Motion Blur:* It simulates linear motion along a direction $\theta$ with the extent of this motion defined by length $l$. It is simulated by using the MATLAB function *fspecial("motion")*, which takes length $l$ and angle $\theta$ as input and generates the motion blur kernel.

In addition, to mimic the localized distortions typically found in real multispectral blurred images, we generate binary masks of different shapes and sizes. The localized blurred image is finally obtained by blending the clean image with the globally blurred image using the binary masks. This procedure is separately applied to each band of the multispectral (MS) image and the panchromatic (PAN) image to obtain the localized blurred images. The parameters used for the blur simulation for three distortion levels are mentioned in Table III.

TABLE III
PARAMETER VALUES USED FOR SIMULATING BLUR. GAUSSIAN BLUR IS DEFINED USING STANDARD DEVIATION $\sigma$, MOTION BLUR USES LENGTH $l$ AND ANGLE $\theta$ AS PARAMETERS, AND DEFOCUS BLUR IS PARAMETERIZED BY RADIUS $r$

| Parameter | Level-1 (Low) | Level-2 (Moderate) | Level-3 (High) |
|---|---|---|---|
| $\sigma$ | $0.5 - 1$ | $0.9 - 3$ | $2 - 5$ |
| $l$, $\theta$ | $3 - 5$ , $30$ | $5 - 11$ , $30$ | $11 - 17$ , $30$ |
| $r$ | $1$ | $1.2 - 2.5$ | $2 - 5$ |

TABLE IV
PARAMETER VALUES USED FOR SIMULATING STRIPE NOISE

| Parameter | Level-1 (Low) | Level-2 (Moderate) | Level-3 (High) |
|---|---|---|---|
| $I$ (offset) | $(-10, 10)$ | $(-14, 14)$ | $(-20, 20)$ |

*4) Stripes:* Stripe noise can be modeled using the equation $y(t) = x(t) + s(t)$, where $y(t)$ represents the striped image, $x(t)$ corresponds to the clean image, and $s(t)$ is the stripe component. We simulate periodic and nonperiodic stripe noise patterns based on the offset-based stripe generation model described in [57] and [58]. Specifically, for a particular distortion level, one of the two given values described in Table IV is randomly chosen as the stripe offset value represented by $I$. The values correspond to the case when the intensity is in the range $[0, 255]$. The stripe component $s(t)$ is generated using periodically spaced or randomly spaced (nonperiodic) lines of intensity $I$ with a fixed random angle in $(-10°, 10°)$ from the vertical axis for each image. The stripes are oriented close to vertical, resembling typical distortions observed in real images.

*5) Haze:* We simulate nonhomogeneous haze on multispectral and panchromatic images as an additive model, as proposed in [59], [60], and [61]. We extract the haze thickness map (HTM), as described in [60] and [61], from the hazy images of Cartosat and WorldView sensors. We add this HTM to the blue band of the multispectral image for multispectral haze simulation. For other bands, we simulate haze by following the Rayleigh scattering model using the HTM obtained for the blue band, as described in [60] and [61]. The choice of different models for different bands stems from the prominence of haze signatures in the blue band due to its shorter wavelength, whereas direct HTM extraction from other bands often introduces structural artifacts or noise. Thus, the blue band provides a more stable and consistent basis for haze simulation across bands. For panchromatic images, we add the obtained HTM to the panchromatic image. To vary the haze level among the images, we multiply a parameter $\beta$ with HTM. The parameter $\beta$ is selected from the set $\{1.10, 1.05, 0.95, 0.85\}$.

A sample set of simulated distortions is shown in Fig. 3. Table V presents an overview of our dataset that includes both real and simulated distortions.

### C. Subjective Study

Many subjective studies for the QA of natural images involve rating images on a continuous scale with a single stim-

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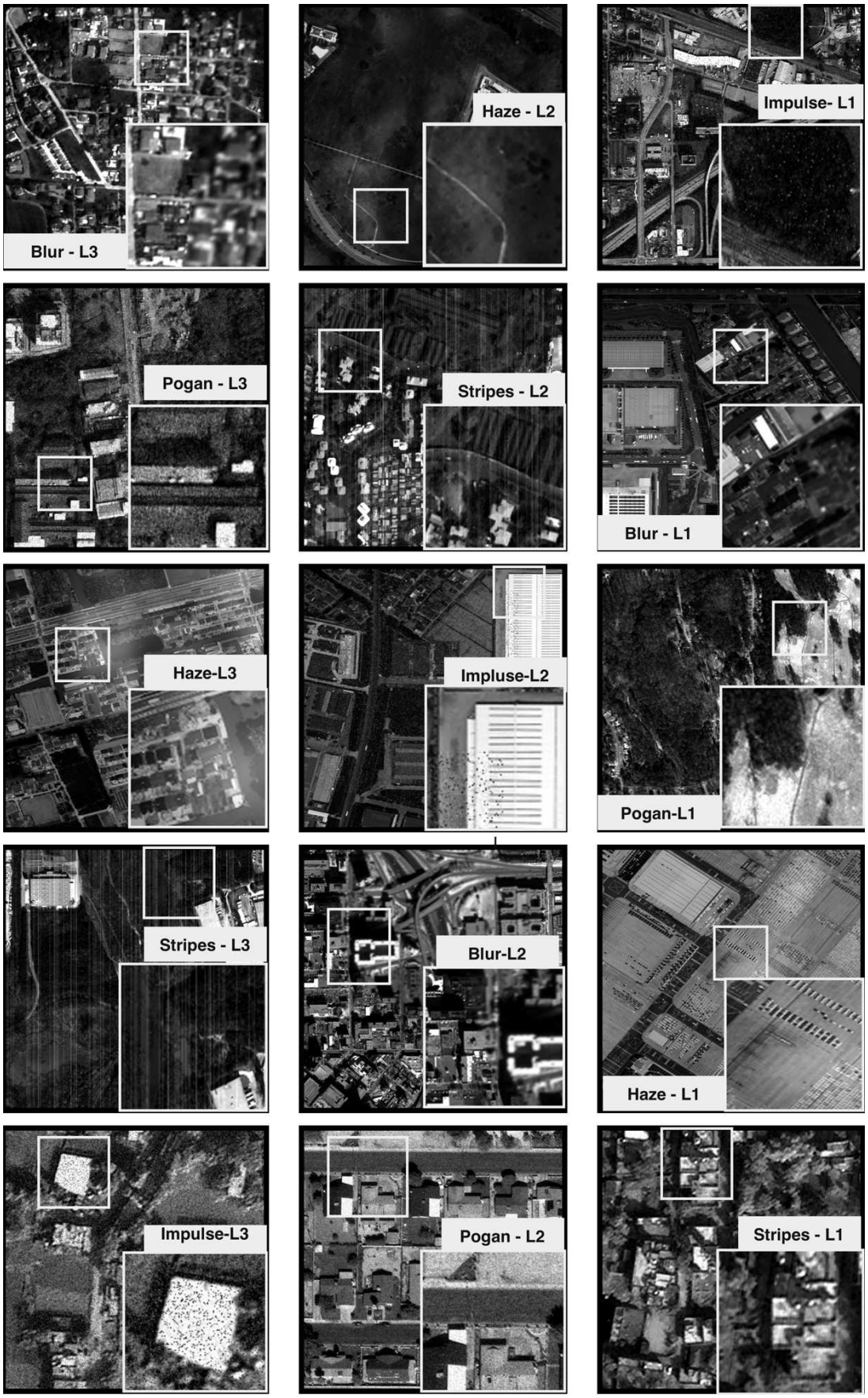



Fig. 3. Examples of simulated distortions in the MRSID dataset. The image grid follows the same structure as described in Fig. 2.

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TABLE V
MRSID OVERVIEW; "SIM." CORRESPONDS TO SIMULATED DISTORTIONS

| Multispectral | | | | Panchromatic | | | |
|---|---|---|---|---|---|---|---|
| Cartosat-3 | | Worldview-2/3 | | Cartosat-3 | | Worldview-2/3 | |
| Real | Sim. | Real | Sim. | Real | Sim. | Real | Sim. |
| 380 | 120 | 56 | 177 | 158 | 57 | 20 | 133 |
| Total – 500 | | Total – 233 | | Total – 215 | | Total – 153 | |
| Total (Multispectral): 733 | | | | Total (Panchromatic): 368 | | | |
| Total Images in Database: 1101 | | | | | | | |

ulus to obtain an MOS for each image [62], [63], [64], [65], [66]. However, as human subjects are less familiar with diverse satellite image content and the distortions are often localized for remote sensing images, we believe that a continuous scale-based subjective study may not be the most appropriate and reliable. Thus, we opt for a pairwise subjective study to collect more reliable subjective opinions. Pairwise subjective studies are popular in IQA as humans exhibit higher consistency in relative judgments than in assigning absolute scores. Seminal datasets and methods [67], [68], [69] show that pairwise preference data yields robust, reproducible quality scales via Bradley–Terry/Elo-style psychometric mapping. Moreover, recent works [70], [71] argue that preference-based supervision directly addresses content- and observer-dependent variability, offering stable scaling from sparse comparisons and improving downstream IQA learning. The International Telecommunication Union (ITU) also recommends pairwise comparison tests for subjective QA [72].

In our pairwise subjective study, we display a pair of images to the subjects, and each subject selects the image with the higher quality. We generate 3000 image pairs from 1101 images in MRSID, of which 2000 are multispectral pairs and 1000 are panchromatic pairs. Among the 2000 multispectral pairs, we generate 500 pairs per band. Note that a pair of images is formed from the same multispectral band or panchromatic images.

We generate pairs as follows. We segregate images in our dataset based on our coarse annotation of the type and level of distortions. We randomly generate pairs from within distortions (e.g., haze versus haze) and across distortions (e.g., noise versus blur) with equal probability. Furthermore, in each case, we select pairs with the same level of distortions with a probability of 0.2, adjacent levels with a probability of 0.5, and pairs with one image having the lowest and another with the highest level of distortion with a probability of 0.3.

*1) Study Details:* We conduct a subjective study with 62 subjects in which each subject participates in one or at most two sessions with a gap of at least 24 h per session. No subject is shown the same set of images twice. Each subject rates around 167 pairs of images in a controlled laboratory environment on the same computer system (Intel Core i7-6700, 4.0 GHz, eight cores, 32-GB RAM) by viewing images on the same monitor (24-in Dell, $1920 \times 1080$ resolution). Prior to each session, a training phase is also conducted where 10–15 sample image pairs are shown to users, allowing them to familiarize themselves with the framework and gain an understanding of the diverse distortions present in the study. Fig. 4 illustrates the interface used for the subjective study and is developed in-house using Python GUI-supported tools such as *Tkinter*. The training pairs are excluded from the image pairs that we use in the final assessment. All the subjects are from the student community of the Indian Institute of Science, and informed consent from every subject is obtained before the study.

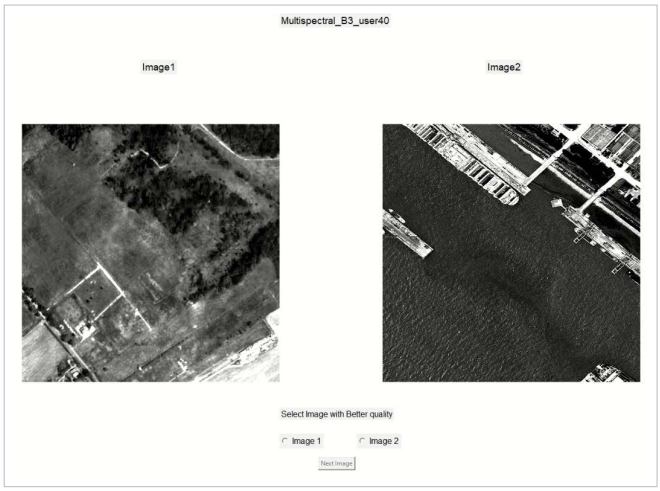



Fig. 4. Screenshot of our study interface.

*2) Reliability of Subjective Scores:* The first phase of the subjective study was conducted with 55 subjects out of 62 subjects to get five ratings per image pair. Based on these ratings, we observe that, in most pairs, an image is rated superior at least four out of five times; in a few pairs, it is selected three out of five times. We refer to the samples that received three out of five votes as ambiguous pairs. To make the pairwise rankings more robust for the ambiguous pairs, we get two more ratings for them, resulting in seven ratings for these ambiguous pairs. Seven additional subjects participate in annotating these ambiguous pairs. The numbers of ambiguous pairs in the blue, green, red, and NIR bands are 130, 155, 126, and 137, respectively, each out of 500 pairs, and 211 in 1000 panchromatic pairs, totaling 759 ambiguous pairs out of 3000 pairs annotated by seven subjects. Thus, in total, we collect 16 518 human annotations in our study.

To evaluate the reliability of our study, we check for consistency and identify the number of pairs where the image rated superior in quality does not change when either five or seven ratings are used. Note that image pairs that already had at least four out of five ratings in favor of one image are also included in this comparison since even if we collect two more ratings, the winner does not change. On collecting the additional ratings for these pairs, 2800 out of 3000 pairs are consistent. This leads to an overall consistency of 93.33% in the collected ratings, showing that our subjective study is reliable.

*3) Dataset Distortion Analysis Using Subjective Rankings:* To better understand the perceptual quality of various distortions in our dataset, we analyze pairwise annotations where

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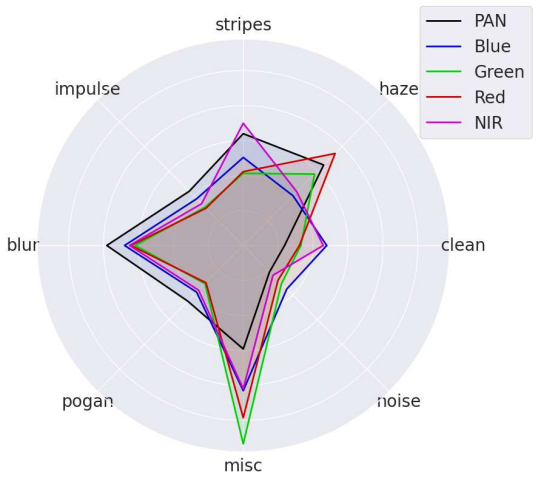



Fig. 5. Fraction of images present for each distortion type in each band.

each pair contains two images with different distortions. We compute the worse ratio for a given distortion as the fraction of times an image with a particular distortion has poorer quality compared to the other image in the pair. We conduct this analysis separately for the real distortions and the simulated distortions. The analysis from Table VI shows clear trends in human perceptual judgments.

1) Noise and stripe distortions are among the most perceptually unpleasant across both real and simulated sources.
2) Distortions such as blur and haze show considerable variation in losing ratios between simulated and real domains, possibly due to differences in their severity or how realistically they are rendered.

This analysis highlights how human perception varies across different distortion types. The distortions analyzed here correspond to those described in Section III-B. Simulated distortions include blur, stripes, impulse noise, pogan, and haze, and real distortions include blur, haze, noise, stripes, and a miscellaneous (misc.) category. The miscellaneous category encompasses diverse artifacts such as banding, butting zone artifacts, and saturation.

We analyze the distribution of real distortions by counting the number of images corresponding to each distortion type within every spectral band. Fig. 5 illustrates the fraction of real distortions across spectral bands. We observe that while the miscellaneous category contains a large number of diverse artifacts, blur is the most prominent specific distortion across all bands, followed by haze and stripes in certain bands. Noise is the least perceived distortion across all bands. Note that synthetic distortions are sampled uniformly across all types.

## IV. Benchmarking Existing Image Quality Measures

We evaluate several existing no-reference IQA (NR-IQA) models on our dataset under different training configurations.

### A. Experimental Setup

We describe two key settings to benchmark the NR-IQA models on MRSID.

*1) Training With Multispectral Bands:* We first define two experimental settings based on how we utilize different bands to train the benchmark methods.

*a) Single-band setting:* We have separate models trained and evaluated for each spectral band, i.e., the model trained on PAN is tested on PAN, the model trained on the blue band is tested on the blue band, and so on through to the NIR band. This setup ensures that each model learns representations specific to the corresponding band and allows us to measure the band-level performance of various NR-IQA methods.

*b) Multiband setting:* In a multiband setting, a single model is trained on the combined training data from all five bands. After training, this unified model is evaluated on the test set of each individual band. This setup allows the model to potentially learn band-invariant quality features and transfer knowledge across bands.

In our experimental design, the test data used for both settings are exactly the same. The only change is in the train data, where, in the single-band setting, models are trained on individual bands, and in the multiband setting, the model is trained on all the bands put together. Despite the multispectral data being dissimilar across bands, we show in Section IV-D that training jointly on multispectral bands outperforms individually trained models.

*2) Data Split Settings:* We evaluate both the above models on two main evaluation settings.

*a) Random split:* We create the train–test splits by randomly splitting all the human-rated pairs into the two sets roughly in the ratio 80:20. We ensure that none of the images in pairs in the train set are part of the test set pairs. We create 15 such splits for evaluation.

*b) Cross-scene split:* In the second setting, we create the train–test split using disjoint sets of areas imaged by the satellite to assess the generalizability of the models. This approach determines whether a model trained on images from certain regions can accurately predict quality when tested on unseen regions, which is an important attribute of a good QA model. For multispectral bands, we use six regions (three from Cartosat and three from WorldView: Shanghai, Atlanta, and Vegas). We use pairs containing images from any four regions for the training set and pairs from the remaining two regions for the testing set. We create five such combinations of train–test splits.

For panchromatic images, we use four regions (one from Cartosat and the three from WorldView: Shanghai, Khartoum, and Vegas). Pairs containing images from any two regions are used for the training set, and pairs from the remaining two regions are used for the testing set, resulting in three combinations of train–test splits.

*3) Evaluation Measure:* To evaluate the performance of our model, we use the accuracy of predicting the correct rank based on pairwise comparisons. Each data point consists of a pair of images, the accuracy is computed as the fraction of times, and the correct rank was predicted with respect to the ground-truth human rankings.

*a) Random split:* The final accuracy reported is the median of the accuracy, computed as described above, across all 15 random splits.

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*b) Cross-scene split:* This setting is designed to test generalization across different scenes. For the multispectral bands, there are five scene-based splits; for the PAN band, there are three splits. Since selecting pairs from the same region in each test split can sometimes result in very few test pairs, we evaluate the model performance by aggregating the evaluations across different splits. Let $i \in \{1, 2, \ldots, S\}$ index the split where $S$ is the number of splits. In each split, let the number of correctly predicted pairs be $C_i$ and the total number of pairs be $T_i$. The overall accuracy is then calculated as

$$\text{Accuracy} = \frac{\sum_{i=1}^{S} C_i}{\sum_{i=1}^{S} T_i}.$$

Note that, in the literature, the pairwise comparisons are sometimes converted to absolute scores [67], [68]; however, this can still be a noisy conversion. Thus, we directly evaluate the pairs with the ground truth using the pairwise accuracy metric. This choice is consistent with recent works that adopt pairwise accuracy for QA [73], [74], [75], [76].

### B. Training on Pairwise Data

To fine-tune the models based on the pairwise rankings available in the database, we follow the approach of [77] and employ a pairwise ranking loss. We use pairwise supervision where each data sample consists of a pair of images $(I_1, I_2)$ along with a binary label $y \in \{0, 1\}$. The label indicates which image has better perceptual quality: $y = 0$ implies that $I_1$ is better than $I_2$, and $y = 1$ indicates the opposite. The models predict a scalar quality score for each image denoted as $f(I_1)$ and $f(I_2)$. During training, we compute the sigmoid of the difference between the predicted scores as $\hat{y} = \sigma(f(I_2) - f(I_1))$, where $\sigma(\cdot)$ is the sigmoid function. We use the binary cross-entropy (BCE) loss to optimize the model. The loss is defined as

$$\mathcal{L} = -y \log \hat{y}.$$

The above loss is averaged over all image pairs in a batch to obtain $\mathcal{L}_{\text{BCE}}$.

### C. Benchmarked Methods

We benchmark the performance of several classical and deep learning-based NR-IQA methods. Classical methods include BRISQUE [14], NIQE [16], HOSA [31], and IL-NIQE [32], while deep learning-based methods include DBCNN [17], HyperIQA [18], TReS [20], MetaIQA [24], MUSIQ [23], MANIQA [19], ReIQA [21], and Q-Former [22]. To evaluate the performance of the models, we compute pairwise accuracy in selecting the higher quality image by passing both images in a pair through the model to predict the image with higher quality.

We now discuss how each of the models described above is fine-tuned using the pairwise data. To fine-tune BRISQUE [14] on our dataset, we extract the 36-D feature vector for each image and feed this vector to a neural network-based regressor model. We adopt a similar benchmarking process for HOSA [31], where a 14 700-D feature vector is extracted from each image. For NIQE [16] and IL-NIQE [32], we use a pristine set of around 55–60 remote sensing images in each band, which are not a part of our database to estimate the pristine model parameters. In particular, we sampled from the Cartosat and Worldiew scenes of high quality based on visual inspection to obtain the pristine set of images.

TABLE VI

DISTORTIONWISE WORSE RATIOS ($R = W/T$), WHERE $W$ IS THE NUMBER OF TIMES AN IMAGE WAS RATED WORSE AND $T$ IS THE TOTAL NUMBER OF APPEARANCES

| Simulated Distortions | | | | Real Distortions | | | |
|---|---|---|---|---|---|---|---|
| **Type** | $W$ | $T$ | $R$ | **Type** | $W$ | $T$ | $R$ |
| blur | 162 | 255 | 0.635 | stripes | 67 | 98 | 0.684 |
| stripes | 189 | 325 | 0.582 | misc. | 277 | 427 | 0.649 |
| impulse | 84 | 160 | 0.525 | noise | 120 | 193 | 0.622 |
| pogan | 82 | 157 | 0.522 | haze | 104 | 197 | 0.528 |
| haze | 57 | 251 | 0.227 | blur | 120 | 252 | 0.476 |

Among deep NR-IQA methods, we fine-tune HyperIQA [18], TReS [20], MetaIQA [24], and MUSIQ [23] end-to-end. We fine-tune only the last fully connected layer for DBCNN [17] since the earlier layers are pretrained with specific objectives. Similarly, in MANIQA [19], we fine-tune only the score and weight branches of the network. This selective fine-tuning strategy is adopted because end-to-end training resulted in poorer performance compared to partial fine-tuning, likely due to overfitting on the limited training data. In ReIQA [21], we only fine-tune the final layer in the quality-aware encoder and the quality regression network due to resource constraints. In Q-Former [22], we use a BLIP-2-based [78] architecture for fine-tuning. Furthermore, we keep the vision encoder frozen, and only the query tokens and the Q-Former module are trainable. We regress the averaged Q-Former output tokens through a linear layer to predict quality.

### D. Performance Analysis

The results of single-band training and multiband training on panchromatic and multispectral bands for both random splits and cross-scene splits are shown in Tables VII–IX. Tables VII and IX provide the accuracy of predicting the correct rank from a pair of images. We show both the accuracies for individual spectral bands and an aggregate performance across all bands. The benchmarking results for both settings demonstrate that multiband training enhances the performance of various nondeep and deep NR-IQA methods compared to single-band training. This indicates that leveraging information across multiple spectral bands enables models to better generalize and capture subtle variations in image quality, which are not always apparent within individual bands during training. Furthermore, it is evident that deep learning methods surpass nondeep learning methods across most bands, particularly after fine-tuning, indicating that deep learning models are more adept at capturing the intricate quality variations present in remote sensing images.

In the random split benchmarking, in nondeep NR-IQA methods, HOSA emerges as the model that benefits most from multiband training. Although other methods such as NIQE and IL-NIQE show minor gains, their improvements are inconsistent across bands. In the deep NR-IQA category, Q-Former shows the most significant boost due to multiband training. It also maintains leading performance across all bands

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TABLE VII
BENCHMARKING RESULTS FOR NR-IQA METHODS (RANDOM SPLIT)

| Method | Single Band Training Accuracy (in %) | | | | | | Multi Band Training Accuracy (in %) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PAN | Blue | Green | Red | NIR | Avg | PAN | Blue | Green | Red | NIR | Avg |
| **Non Deep NR-IQA** | | | | | | | | | | | | |
| BRISQUE | 71.66 | 70.17 | 70.68 | 67.24 | 69.09 | 69.90 | 77.58 | 70.90 | 63.15 | 64.51 | 65.62 | 68.89 |
| NIQE | 66.07 | 68.25 | 72.41 | 71.64 | 67.14 | 68.77 | 63.73 | 67.74 | 67.79 | 66.07 | 66.67 | 66.67 |
| HOSA | 80.90 | 72.13 | 76.67 | 69.84 | 66.66 | 72.99 | 78.84 | 76.19 | 77.58 | 79.03 | 75.47 | 77.55 |
| IL NIQE | 70.67 | 67.74 | 61.01 | 63.63 | 61.81 | 65.67 | 62.03 | 66.59 | 69.75 | 66.23 | 63.66 | 66.38 |
| **Deep NR-IQA** | | | | | | | | | | | | |
| DBCNN | 81.30 | 69.64 | 69.64 | 71.87 | 68.85 | 71.04 | 82.46 | 79.25 | 73.68 | 75.81 | 73.58 | 77.94 |
| HyperIQA | 79.23 | 77.42 | 69.70 | 71.19 | 71.43 | 72.05 | 82.41 | 78.95 | 75.86 | 79.31 | 74.07 | 77.81 |
| TReS | 82.45 | 81.82 | 73.77 | 75.00 | 70.49 | 75.07 | 82.66 | 81.88 | 79.13 | 80.48 | 77.10 | 80.49 |
| MetaIQA | 77.31 | 73.68 | 72.58 | 71.64 | 65.51 | 70.62 | 76.56 | 79.69 | 74.24 | 74.60 | 65.62 | 74.84 |
| MANIQA | 77.19 | 71.93 | 68.33 | 72.58 | 69.24 | 70.57 | 73.28 | 69.35 | 63.79 | 67.74 | 64.15 | 67.74 |
| MUSIQ | 77.68 | 73.21 | 69.64 | 67.86 | 66.07 | 69.67 | 79.00 | 79.00 | 73.00 | 77.00 | 70.00 | 75.97 |
| CLIPIQA+ | 53.33 | 56.45 | 50.82 | 50.00 | 55.38 | 53.18 | 48.78 | 50.00 | 47.46 | 52.54 | 42.59 | 48.39 |
| ReIQA | **82.35** | **83.60** | **77.97** | **77.42** | 71.70 | **77.79** | 83.19 | 85.00 | 80.36 | 81.07 | 76.25 | 81.83 |
| Q-Former | 81.03 | 76.56 | 72.73 | 74.19 | **73.44** | 76.20 | **91.82** | **91.67** | **91.94** | **88.71** | **92.31** | **91.23** |

TABLE VIII
MEDIAN SROCC VALUES OF LEADING METHODS UNDER RANDOM-SPLIT MULTIBAND TRAINING

| Method | PAN | Blue | Green | Red | NIR | Avg |
|---|---|---|---|---|---|---|
| TReS | 0.744 | 0.798 | 0.770 | 0.779 | 0.766 | 0.767 |
| ReIQA | 0.686 | 0.679 | 0.670 | 0.702 | 0.567 | 0.665 |
| Q-Former | **0.852** | **0.838** | **0.862** | **0.863** | **0.887** | **0.859** |

in the multiband setting, outperforming all other methods. We also note that both TReS and ReIQA also benefit substantially from multiband training.

In the cross-scene split benchmarking, HOSA again stands out among nondeep NR-IQA methods. Its performance indicates a strong ability to generalize well to unseen scenes. BRISQUE and IL-NIQE show moderate improvements but not as pronounced or consistent across all bands. Among the Deep NR-IQA methods, Q-Former still remains the top performer, benefiting from multiband training. It shows a significant increase from its single-band performance. It also shows consistent high accuracy across all bands, highlighting its superior ability to leverage multiband training. TReS also benefits heavily, in the multiband scenario compared to single-band training.

To further assess the correlation of our model predictions with human perceptual judgments, we convert the pairwise comparisons into image-level subjective scores using the Elo rating system [79], consistent with prior IQA datasets that derive absolute scores from pairwise data [68], [69]. We compute the Spearman rank-order correlation coefficient (SROCC) between model predictions and image-level subjective scores for the top three methods (TReS, ReIQA, and our Q-Former) under the random-split multiband training setup, as shown in Table VIII. Similar to the pairwise accuracy observations in Table VII, our model demonstrates superior performance consistently across all spectral bands.

We arrive at the following key takeaways from the benchmark results.

1) Q-Former benefits the most from multiband training, consistently outperforming all other models across both settings and all bands. This highlights its strong capacity for learning cross-band quality features.
2) HOSA shows the most improvement among nondeep methods, especially in red and NIR bands, indicating its relative robustness when extended to multiband inputs.
3) Deep models demonstrate greater benefits from multiband training compared to nondeep methods, showcasing their superior ability to integrate spectral information and learn complex interband dependencies.

### E. Impact of Fine-Tuning on Multispectral Images

We now analyze the impact of fine-tuning the models on multispectral images as opposed to directly using the

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TABLE IX
BENCHMARKING RESULTS FOR NR-IQA METHODS (CROSS-SCENE SPLIT)

| Method | Single Band Training Accuracy (in %) | | | | | | Multi Band Training Accuracy (in %) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PAN | Blue | Green | Red | NIR | Avg | PAN | Blue | Green | Red | NIR | Avg |
| **Non Deep NR-IQA** | | | | | | | | | | | | |
| BRISQUE | 71.56 | 73.72 | 70.81 | 61.09 | 67.19 | 68.59 | 63.23 | 72.26 | 64.91 | 68.17 | 63.63 | 66.66 |
| NIQE | 64.21 | 72.26 | 69.83 | 71.70 | 66.40 | 69.26 | 56.84 | 73.36 | 64.92 | 65.27 | 63.24 | 66.67 |
| HOSA | 68.62 | 75.91 | 70.49 | 62.37 | 67.19 | 68.81 | 76.96 | 81.75 | 86.89 | 79.74 | 84.98 | 82.33 |
| IL NIQE | 65.19 | 71.16 | 62.29 | 64.30 | 67.58 | 65.99 | 57.84 | 73.36 | 67.54 | 71.06 | 67.19 | 68.00 |
| **Deep NR-IQA** | | | | | | | | | | | | |
| DBCNN | 69.61 | 70.07 | 75.37 | 71.70 | 63.24 | 70.29 | 80.88 | 76.28 | 82.62 | 75.24 | 70.75 | 71.30 |
| HyperIQA | 65.20 | 77.37 | 71.15 | 66.56 | 69.56 | 70.16 | 70.59 | 79.56 | 79.34 | 81.35 | 81.42 | 78.91 |
| TReS | 70.59 | 76.64 | 74.75 | 72.35 | 68.77 | 72.83 | 75.00 | 84.31 | 83.28 | 83.92 | 85.77 | 82.25 |
| MetaIQA | 64.71 | 72.99 | 54.75 | 68.81 | 73.52 | 66.74 | 66.67 | 74.82 | 71.48 | 73.31 | 67.78 | 71.20 |
| MANIQA | 66.66 | 75.91 | 69.50 | 71.06 | 75.09 | 71.78 | 63.73 | 69.71 | 63.28 | 69.77 | 65.22 | 66.52 |
| MUSIQ | 65.19 | 80.29 | 64.26 | 66.23 | 67.19 | 68.67 | 67.00 | 83.00 | 79.00 | 81.00 | 80.00 | 78.65 |
| CLIPIQA+ | 45.58 | 52.19 | 54.42 | 50.16 | 53.75 | 51.75 | 52.45 | 54.74 | 48.52 | 53.05 | 50.20 | 51.83 |
| ReIQA | **75.49** | 80.29 | **78.36** | **74.28** | 69.96 | **75.80** | 76.68 | 81.76 | 79.93 | 81.64 | 83.28 | 80.76 |
| Q-Former | 69.60 | **81.38** | 66.22 | 70.09 | **73.91** | 72.37 | **86.76** | **89.42** | **92.13** | **83.28** | **89.33** | **88.20** |

TABLE X
PRETRAINED (PT) AND FINE-TUNED (FT) AVERAGE ACCURACIES (%) ACROSS ALL BANDS

| Method | Random Split | | Cross-Scene Split | |
|---|---|---|---|---|
| | PT | FT | PT | FT |
| BRISQUE | 59.98 | **69.90** | 59.16 | **68.59** |
| NIQE | 63.73 | **68.77** | 65.03 | **69.26** |
| HOSA | 62.50 | **72.99** | 61.76 | **68.81** |
| IL NIQE | 60.62 | **65.67** | 59.31 | **65.99** |
| DBCNN | 48.06 | **71.04** | 44.17 | **70.29** |
| HyperIQA | 58.94 | **72.05** | 58.20 | **70.16** |
| TReS | 52.85 | **75.07** | 53.53 | **72.83** |
| MetaIQA | 51.04 | **70.62** | 49.44 | **66.74** |
| MANIQA | 64.15 | **70.57** | 61.76 | **71.78** |
| MUSIQ | 60.49 | **69.67** | 62.29 | **68.67** |
| ReIQA | **78.61** | 77.79 | **76.76** | 75.80 |
| Q-Former | 49.16 | **76.20** | 49.78 | **72.37** |

pretrained NR-IQA methods. We analyze the aggregated performance across all the spectral bands. The results for both settings in Table X demonstrate that fine-tuning improves the performance of various NR-IQA methods. However, ReIQA lags behind other models despite fine-tuning. This can be attributed to two main reasons. Although ReIQA includes both content- and quality-aware components, only the quality-aware branch is fine-tuned due to resource limitations, limiting its ability to adapt effectively to the target domain. Second, the relatively small size of the fine-tuning dataset, compared to the very large training dataset, further restricts the model's performance.

## V. FURTHER ANALYSIS AND IMPROVEMENT OF Q-FORMER

The benchmarking results in Tables VII and IX indicate that Q-Former consistently outperforms other state-of-the-art NR-IQA models in multiband remote sensing IQA. The consistent superior performance of Q-Former in the multiband setting suggests its enhanced ability to learn generalized quality-aware representations. Thus, we investigate the reasons behind such superior performance. We first discuss how higher alignment of features across different spectral bands contributes to its superior performance. We then present an extension of this model to further increase the interband feature similarity to further improve performance.

### A. Understanding the Superior Performance of Q-Former

To understand why the Q-Former model performs better when trained jointly across bands, we compute the CKA similarity between feature representations from different spectral bands. In particular, the average pooled features of the Q-Former are taken for this analysis. Let the spectral bands be denoted as $B_k$, where $k \in \{1, 2, \ldots, K\}$ and $K$ is the total number of spectral bands. Let $X_i \in \mathbb{R}^{n_1 \times d}$ and $X_j \in \mathbb{R}^{n_2 \times d}$ be the features of two spectral bands concatenated rowwise for

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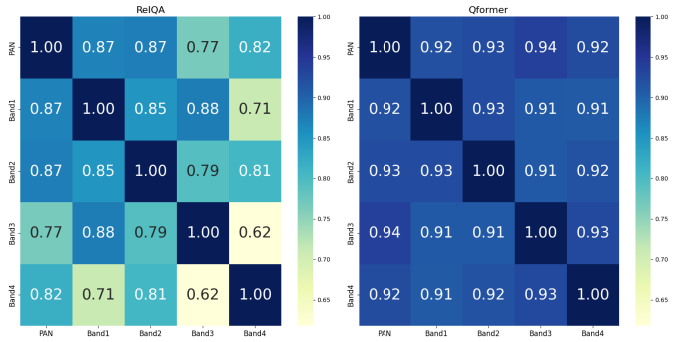



Fig. 6. CKA similarity matrices for (left) ReIQA and (right) Q-Former. Higher values (darker colors) indicate greater similarity between representations of different spectral bands. Q-Former shows higher and more uniform cross-band similarity.

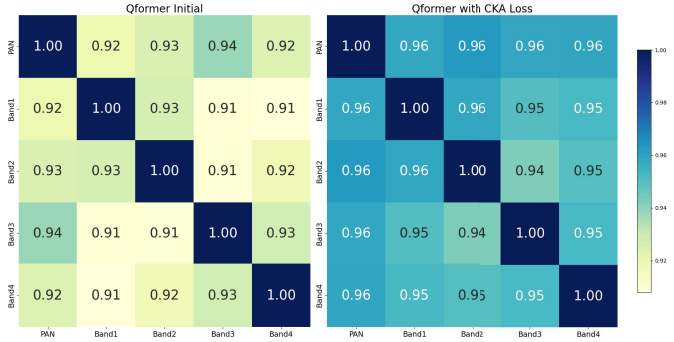



Fig. 7. CKA similarity matrices for (left) Q-Former and (right) Q-Former with CKA loss term. Using the CKA loss term, the CKA similarity has clearly increased.

different samples in band $B_i$ and $B_j$, where $n_1$ and $n_2$ are the number of samples in the respective bands, $d$ is the feature dimension, and $i, j \in \{1, 2, \ldots, K\}$. Linear CKA is defined as

$$\mathrm{CKA}\left(X_i, X_j\right) = \frac{\mathrm{Tr}\left(\left(X_i X_j^\top\right)\left(X_j X_i^\top\right)\right)}{\|X_i X_i^\top\|_F \cdot \|X_j X_j^\top\|_F} \tag{3}$$

where $\|\cdot\|_F$ is the Frobenius norm. The numerator computes the similarity of every sample in $X_i$ with every sample in $X_j$ and sums the squares of the similarity values. The denominator normalizes this similarity with such similarities computed within each band.

We compare the CKA matrix of Q-Former with that of ReIQA in Fig. 6 for all pairs of bands. We see that the CKA matrix for Q-Former has higher cross-band similarities compared to ReIQA. This implies that the Q-Former learns a unified band-invariant feature space. Thus, the Q-Former effectively integrates information across bands, enabling generalization and robustness in multiband scenarios. In contrast, ReIQA shows significantly lower interband CKA scores, implying that its representations are more band-specific and less aligned. Thus, Q-Former's ability to capture a shared latent space across diverse spectral inputs is the key to its superior performance in joint training settings.

### B. Promoting Cross-Band Similarity With CKA Loss

We now further exploit our observation that Q-Former's superior joint training performance stems from its ability to learn a shared latent space across diverse spectral inputs. To explicitly encourage this behavior, we introduce a loss term that promotes higher CKA similarity between representations from different bands. In each iteration, we randomly sample two bands from the five available spectral bands and compute the CKA similarity between their corresponding representations. The CKA loss is defined as

$$\mathcal{L}_{\mathrm{CKA}} = 1 - \mathrm{CKA}\left(X_i, X_j\right) \tag{4}$$

where $\{(i, j) : i, j \in \{1, 2, \ldots, K\}, i \neq j\}$ is randomly sampled in each iteration.

In our training strategy, we alternate between $\mathcal{L}_{\mathrm{BCE}}$ and $\mathcal{L}_{\mathrm{CKA}}$ in a 2:1 ratio over every three iterations. Specifically, the BCE loss is applied in two out of every three iterations to supervise perceptual quality ranking, while the CKA loss is applied in the third iteration to enforce representational alignment across spectral bands. This strategy ensures that the model learns both high-quality pairwise ranking and cross-band generalization through aligned latent spaces.

TABLE XI

AVERAGE ACCURACY (%) ACROSS BANDS OF Q-FORMER BEFORE AND AFTER ADDING THE CKA-BASED LOSS TERM

| Split | Baseline | With $\mathcal{L}_{\mathrm{CKA}}$ |
|---|---|---|
| Random | 90.34 | 90.73 |
| Cross-scene | 88.70 | 91.80 |

We compare the average performance across bands for the random and cross-scene split settings in Table XI. We see that imposing cross-band similarity leads to a performance improvement in both settings. These results show that explicitly aligning representations across bands further improves the model's ability to generalize across bands while training on multiple bands. We also see from Fig. 7 that the CKA similarity on the test set also increases for the model trained with the new loss term.

## VI. CONCLUSION

In this work, we construct the MRSID by collecting real remote sensing images, simulating diverse distortions, and obtaining pairwise human preference judgments. We benchmark several NR-IQA models originally designed for natural images on multispectral images and observe that fine-tuning on MRSID significantly improves performance. Furthermore, deep models benefit more from training a unified model on multiple bands than from training separate models on individual bands. Q-Former especially demonstrates the most consistent and superior performance across multiband settings, highlighting its ability to effectively learn cross-band quality representations. Our analysis further reveals that Q-Former achieves a higher cross-band feature alignment, as indicated by the CKA similarities, than existing methods such as ReIQA. This suggests that Q-Former learns a unified band-invariant feature space that supports better generalization. Finally, we show that explicitly encouraging cross-band similarity using a CKA loss during training further enhances performance and feature consistency, as evidenced by improved CKA scores

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and model accuracy. These results underscore the importance of cross-band representation alignment for robust IQA in multispectral remote sensing.

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